# Plant adaptations: roots, stems and leaves

*9B Plant growth · Lesson 2 of 7*

Date _____ Class _____

## Today I can

- ☐ I can describe what plants need from the soil and the air, and how roots, root hair cells and xylem are adapted to supply water and mineral salts.
- ☐ I can explain how a leaf and its palisade cells are adapted for photosynthesis.

## What happens today

1. Do Now: where do the reactants come from?
2. Learn: What plants need from mineral salts · Roots: collecting water and mineral salts · The route of water through a plant · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 24 to 25 · Workbook pages 17

**What I learned today (one or two sentences)**

# Key Vocabulary

| Term | Definition |
|---|---|
| **mineral salts** | substances dissolved in soil water that plants need to stay healthy, e.g. nitrates |
| **nitrate** | a mineral salt that supplies nitrogen for making proteins |
| **magnesium** | an element plants need to make chlorophyll |
| **potassium** | an element root cells need to absorb water properly |
| **root hair cell** | a root cell with a long extension that gives a large surface area for absorbing water |
| **xylem** | hollow tubes made from dead cells that carry water and mineral salts up the plant |
| **wilt** | to droop because the cells have too little water |
| **palisade cell** | a leaf cell near the upper surface that is packed with chloroplasts |
| **epidermis** | the outer layer of cells of a leaf |
| **cuticle** | the waxy layer on a leaf that stops it losing too much water |
| **surface area** | the total area of the outside of something |

# Quick Review

*where do the reactants come from?*

**1** Write the word equation for photosynthesis.

**2** Which reactant comes from the air, and which comes from the soil?

**3** Name two other things a plant needs that it gets from the soil.

# Core 1 of 2

*Everyone does these. Answer on the card or under it.*

**1** Complete the table to show how plants get what they need for growth. Use the word bank: air, soil water, respiration, photosynthesis, making proteins, making chlorophyll, absorbing water, keeping cells firm.

| Substance | Where it comes from | How the plant uses it |
|---|---|---|
| oxygen | | |
| carbon dioxide | | |
| nitrate salts | | |
| magnesium salts | | |
| potassium salts | | |
| water | | |

**2** Match each part of a leaf to how it helps the plant. Draw a line.

| | |
|---|---|
| palisade cells | carries water and mineral salts from the roots |
| phloem | stops the leaf losing too much water |
| xylem | allow gases to enter and leave the leaf |
| stomata | contain many chloroplasts, near the top where there is most light |
| cuticle | carries dissolved sugars from the leaf to other parts of the plant |

# Core 2 of 2

*Everyone does these. Answer on the card or under it.*

**3** Complete the sentences about the route of water. Use the word bank.

Water enters the plant through _____ cells, which have a large _____ so they absorb water quickly.

The water travels up the root and stem in _____ vessels, which are hollow tubes made from _____ cells.

If a plant has too little water, its cells sag and the plant _____.

# Challenge 1 of 2

*When Core is finished. Full sentences.*

**C1** The Venus flytrap grows in bogs with very few mineral salts in the soil. It catches insects when the trigger hairs in its trap are touched more than once.

**a)** Suggest why a trigger hair has to be touched more than once before the trap shuts.

**b)** State what the Venus flytrap gets from the insects, and explain why it might die without them even though it can photosynthesise.

# Challenge 2 of 2

*When Core is finished. Full sentences.*

**C2** Describe how a leaf is adapted to allow its palisade cells to get all the resources they need for photosynthesis: light, carbon dioxide and water.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** Name the mineral salt a plant needs to make proteins.

**2** How are root hair cells adapted to absorb water?

**3** Why do palisade cells contain so many chloroplasts?

**Today I can:** Goal 1 ☐ Goal 2 ☐